

Data Analytics (DA)

Module-End -1

Module End Assignment:

“Data Cleaning, Analysis & Visualization with Excel”

Assignment Task Overview

Create and analyse an E-commerce dataset in Excel. The assignment should demonstrate the following:

Dataset Link: [E-commerce store dataset](#)

Assignment Task:

- Data entry, import, and basic organization using tables
- Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data
- Remove the column that contains irrelevant data if it exists
- Application of Excel formulas for calculations, summaries, and logical/text operations
- Use the XLOOKUP/VLOOKUP functions to combine the tables.
- Use of Pivot Tables for advanced summarization
- Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak
- Data visualization: create at least three chart types, build an interactive dashboard, and use basic PivotCharts

Deliverables:

- Final Excel file with all steps visible (use multiple sheets or tabs)
- Screenshots illustrating each cleaning/processing step (optional but encouraged)
- One-page summary explaining key findings and decisions in a Word file

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Expected Proficiency
Data Entry & Organization	Spreadsheet Mgmt	Table Creation, Import, Entry	Beginner
Data Cleaning & Transformation	Data Cleaning	Duplicates, Missing/Inconsistent Data	Intermediate
Excel Formulas & Functions	Statistical/Logic	SUM, IF, COUNTIF, TEXT, Average,xlookup/vlookup functions	Intermediate
Descriptive Statistics	Data Analysis	Mean, SD, Mode, Analysis ToolPak	Intermediate
Data Summarization (Pivot Table)	Data Summarization	Creating, Connecting Tables	Intermediate
Data Visualization	Visualization	Bar, Column, Pie, Sparklines, Dashboards	Intermediate
Reporting & Insights	Communication	Clear Findings Summary	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill/Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial credit Marks Range
Data Entry & Table Setup	3	Clean tables; correct import; data types properly formatted	Basic setup, but some improper import/types or minor errors	1 – 2
Data Cleaning & Transformation	5	All cleaning steps correct; missing/inconsistent fixed, combining the tables if necessary; and it is well-documented	Most steps correct; minor errors or incomplete documentation	2 – 4
Excel Formulas & Functions	4	Correct use of SUM, IF, COUNTIF, TEXT; comprehensive application	Some formulas applied, but inconsistent/partial	2 – 3
Descriptive Statistics	4	Accurate calculations; used Analysis ToolPak/similar tool properly	Most stats correct; process partially shown	2 – 3
Data Summarization (Pivot)	3	Created/used Pivot Table to summarize multi-dimensional data	Some fields are summarized; minor or moderate issues	1 – 2
Data Visualization	4	≥3 chart types; clear labels; dashboards interactive/organized	Charts missing/labels unclear/dashboards basic	2-3
Reporting & Insights	2	Clear, concise one-page summary linking findings to visuals	Partial summary, omits some findings or interpretations	1

DA Skill Taxonomy Structure

- Spreadsheet Management
 - Data Entry, Import, Table Creation
- Data Cleaning
 - Duplicates Removal, Missing Data Handling, Consistency Checks
- Formulas & Analysis
 - SUM, IF, COUNTIF, AVERAGE, TEXT Functions, Statistical Measures
- Data Summarization
 - Pivot Tables, Filtering, Sorting
- Visualization
 - Charts, Dashboards, PivotCharts
- Communication
 - Summary, Reporting Key Insights

Evaluation Guidelines

- Open and review all Excel sheets/tabs provided.
- For each section, match student work to the specific skill and sub-skill area in the taxonomy.
- Assign full or partial marks per the rubric, using suggested partial mark ranges for “partially complete” work.
- For ambiguous or incomplete submissions, note what’s missing and correspondingly reduce credit.
- Comments must explain strengths and improvement points for each section.
- Objective assessment: focus on function, correctness, clarity, not on visual design beyond the stated requirements.

Achievement Criteria & Passing Rules

- At least partial (not zero) credit must be received in every section—no component can be left unattempted.
- Students must fully achieve at least two major skills: Data Cleaning & Transformation and Data Visualization.
- Feedback and a detailed scorecard must be given for all evaluated assignments.